

BEYOND NODAL DAMPING

LOCAL TUNING. COLLECTIVE EFFECTS.

Delay-aware PLL co-design: interaction laws and IEEE-39 validation

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1 THE DESIGN QUESTION

How much SG can a grid replace with delayed, locally controlled GFL?

$$P_{\text{GFL}} = \sum_i P_i^0 \rho_i$$

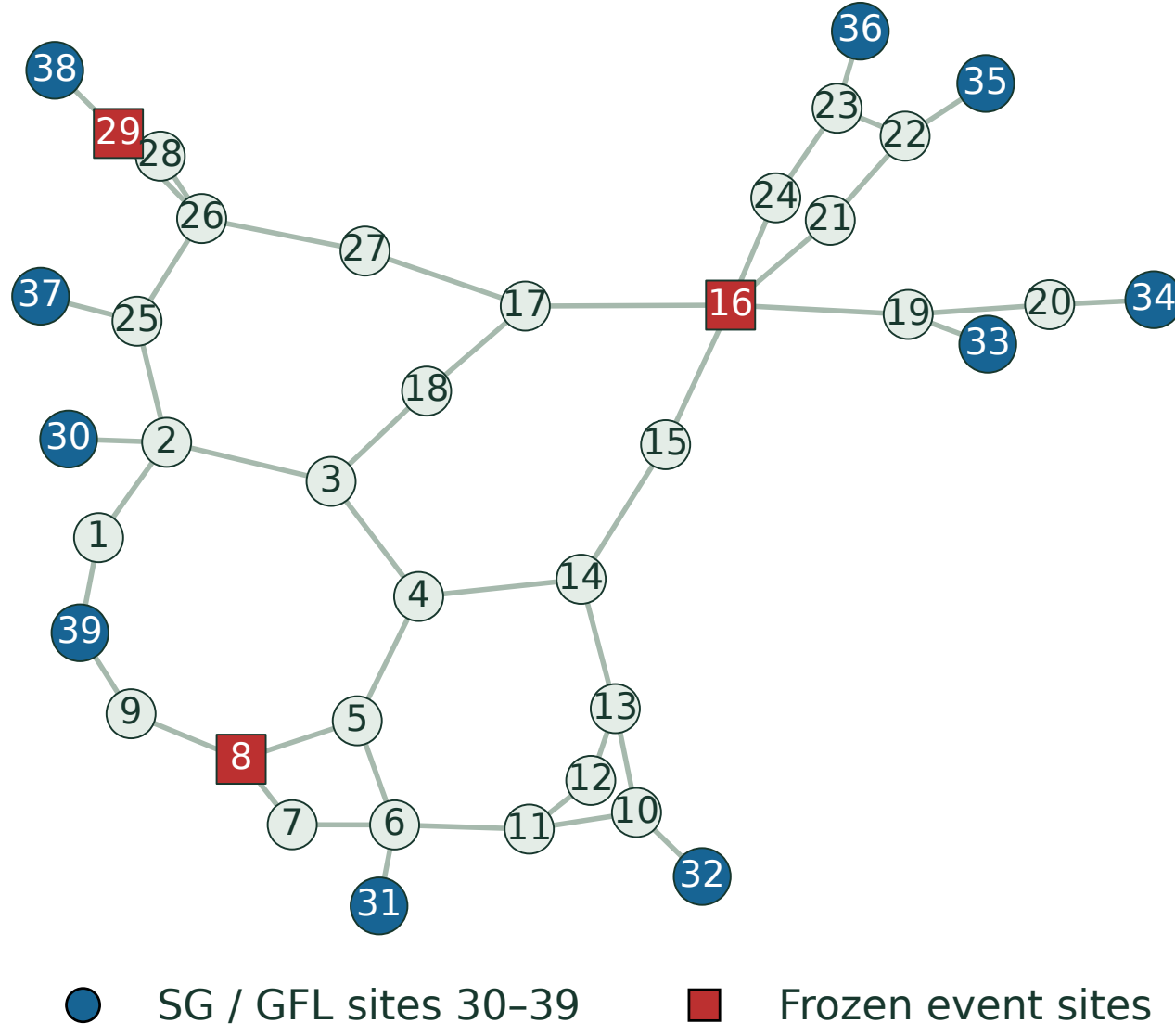
Initialized dispatch P_i^0 ; sharing ρ_i at each generator site.

The physical PLL

$$\begin{aligned} \dot{\vartheta}_i &= \omega_{p,i} \\ t_{f,i} \dot{\omega}_{p,i} &= \xi_i + K_{p,i} e_i^\tau - \omega_{p,i} \\ \dot{\xi}_i &= K_{i,i} e_i^\tau \end{aligned}$$

$e_i^\tau = e_i(t - \tau_i)$ is the delayed voltage phase error. Delays are fixed; only ρ , K_p , K_i are design variables.

2 IEEE-39: A VALIDATED CONSTRUCTION



$$\rho_i : 0.875 \rightarrow 0.876 \quad \Delta P_{\text{GFL}} = 5.403 \text{ MW}$$

A constructed feasible design; replacement was not optimized. Five frozen external events pass the declared nonlinear security checks.

87.6%

GFL dispatch share

4,732.82
MW GFL

669.94
MW retained SG

40 ms
at every PLL

3 GAINS FROM AN EQUATION

Fix one nonreal pole λ and PLL pattern q . Eliminate hidden states:

$$y = G_{\text{PLL}}(\lambda, \rho) q$$

$$W_i = \frac{\lambda^2(1 + t_{f,i}\lambda)e^{\lambda\tau_i}q_i}{y_i}$$

$$K_{p,i} = \frac{\Im W_i}{\Im \lambda}$$

$$K_{i,i} = \Re W_i - \Re \lambda K_{p,i}$$

Requires nonsingular eliminated blocks, $y_i \neq 0$, $\Im \lambda \neq 0$ and admissible gains.

Target: 4.910 Hz. One mode assigned; full spectrum and events need separate checks.

60/60

finite-delay identities
two designs, three frequencies

18–46×

tighter remainder
three pointwise comparisons

5/5

nonlinear events pass
Julia method of steps

485

verified contour panels
one cluster over a full box

4 ONE LOCAL DELAY REDISTRIBUTES COLLECTIVE DAMPING

Physical torque–speed ports

$$\Delta = sI - A_0 - \sum_i b_i c_i^T e^{-s\tau_i}$$

$$Z = M(\Delta_{\nu\nu} - \Delta_{\nu z} \Delta_{zz}^{-1} \Delta_{z\nu})$$

$$D_H(\Omega) = \text{Herm } Z(j\Omega)$$

ν : per-unit SG speed deviation; z : hidden states. $M = \text{diag}[2H_i S_i^0(1 - \rho_i)]$. Harmonic mechanical supply: $\frac{1}{2}\hat{\nu}^* D_H \hat{\nu}$.

Finite-change identity

$$\delta Z = \gamma_i u_i v_i^*$$

$$\delta D_H = \frac{1}{2}(a_i v_i^* + v_i a_i^*)$$

$a_i = \gamma_i u_i$; u_i, v_i are the two network propagation paths from a single PLL channel.

If the paths are independent and $\gamma_i \neq 0$, the change has one positive and one negative eigenvalue.

IEEE-39: +1 ms at one site

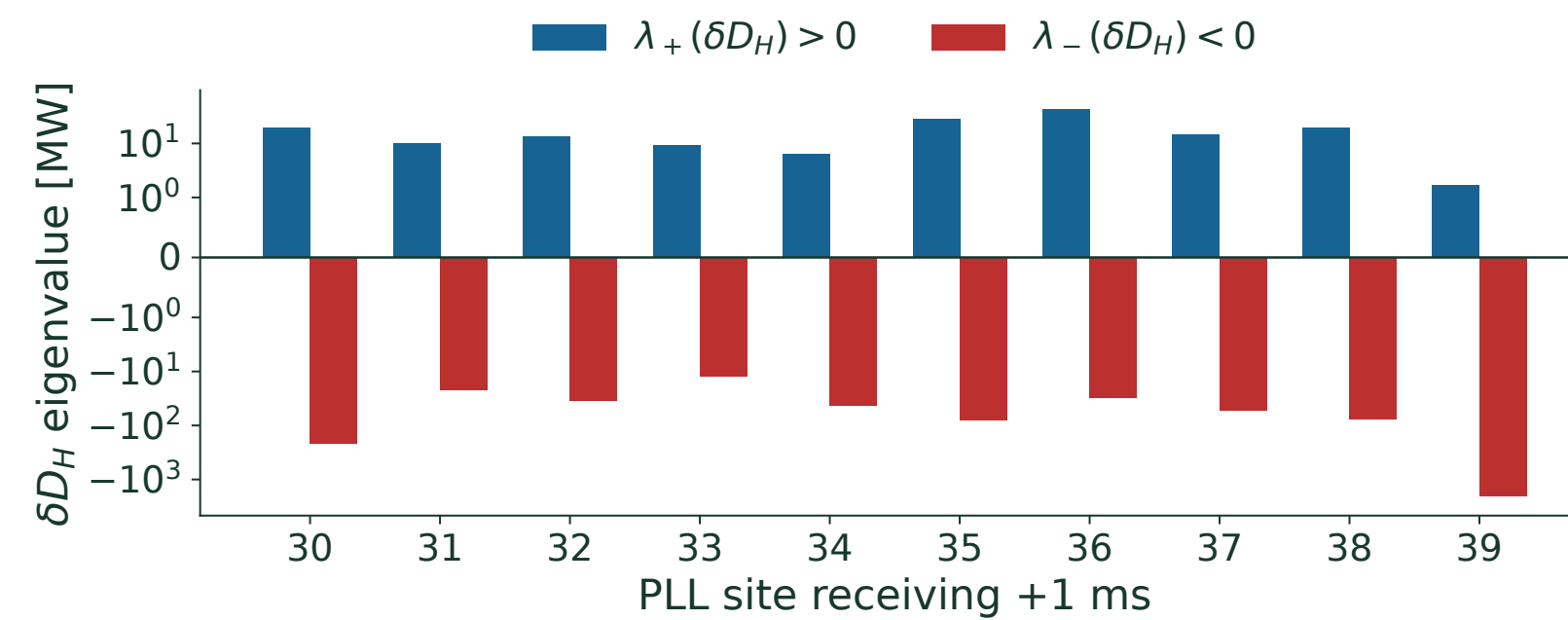


Figure: 5 Hz, analytic design; symmetric-log axis. All 60 checks (2 designs $\times$ 3 frequencies $\times$ 10 sites) pass; error $< 5 \times 10^{-8}$.

Requires invertible hidden blocks. These signs alone do not certify stability.

5 BEYOND ADDING NODAL EFFECTS

Affine descriptor lift $F - F_0 = U\Theta(d)V$; $T = VF_0^{-1}U$ and $H = \Theta(d)T$ on a root contour Γ .

$$z_\Gamma(d) - z_\Gamma(0) = -\frac{1}{2\pi jm} \oint_\Gamma \text{tr} \log(I + H) ds$$

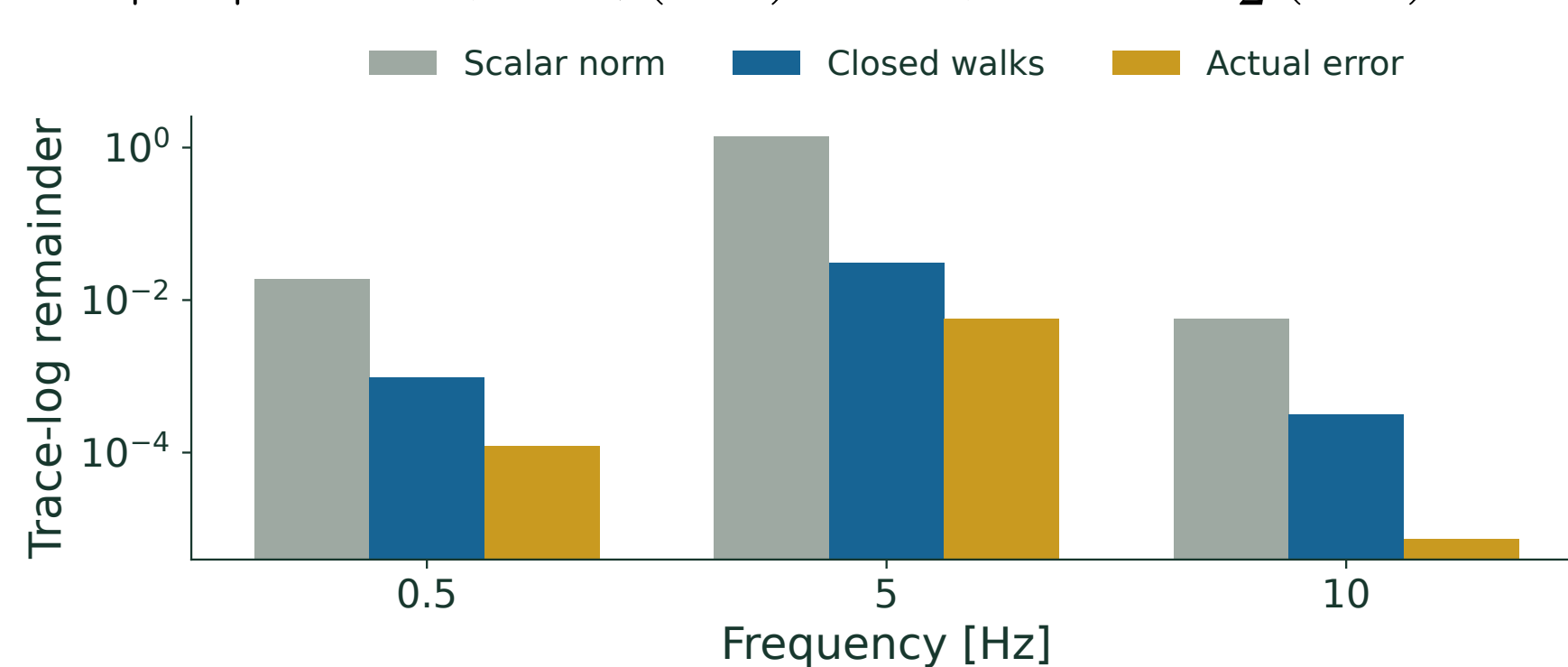
z_Γ : mean of m enclosed poles. A uniform $\|H\| < 1$ preserves the count; the gauge pole is excluded.

$$\mathcal{Q}_{ab} = \Re \frac{1}{2\pi jm} \oint_\Gamma \text{tr}(E_a T E_b T) ds$$

E_a selects a physical parameter. Mixed curvature captures two interventions coupled through return paths. These are signed interactions on the intervention-port graph, not a Laplacian.

6 KEEP THE WALKS, TIGHTEN THE BOUND

$$\|H\| \leq \mathcal{M}, \quad \varrho(\mathcal{M}) < 1, \quad \Psi_2(\mathcal{M}) = -\log \det(I - \mathcal{M}) - \text{tr } \mathcal{M}$$



Three frozen algebraic points; not a replacement-capacity bound. A replacement upper bound additionally needs a verified integral over the complete contour.

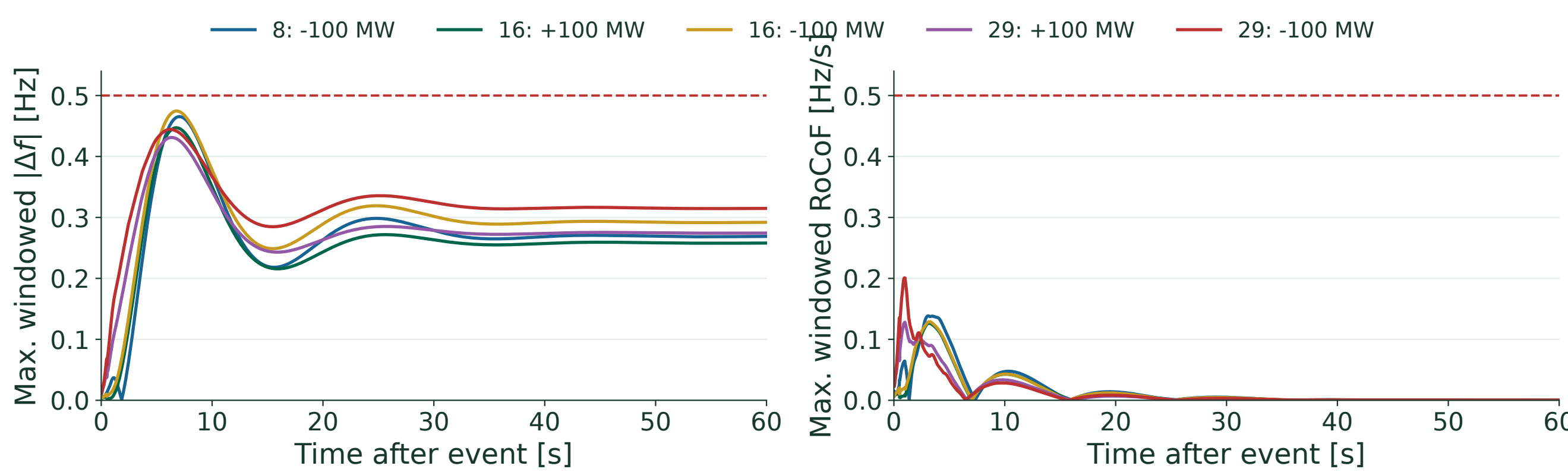
18–46×

smaller remainder bound
than the scalar norm
bound

Same parameter box; heterogeneous delays spanning 0–40 ms. All three points reported.

7 NONLINEAR DELAYED VALIDATION IN JULIA

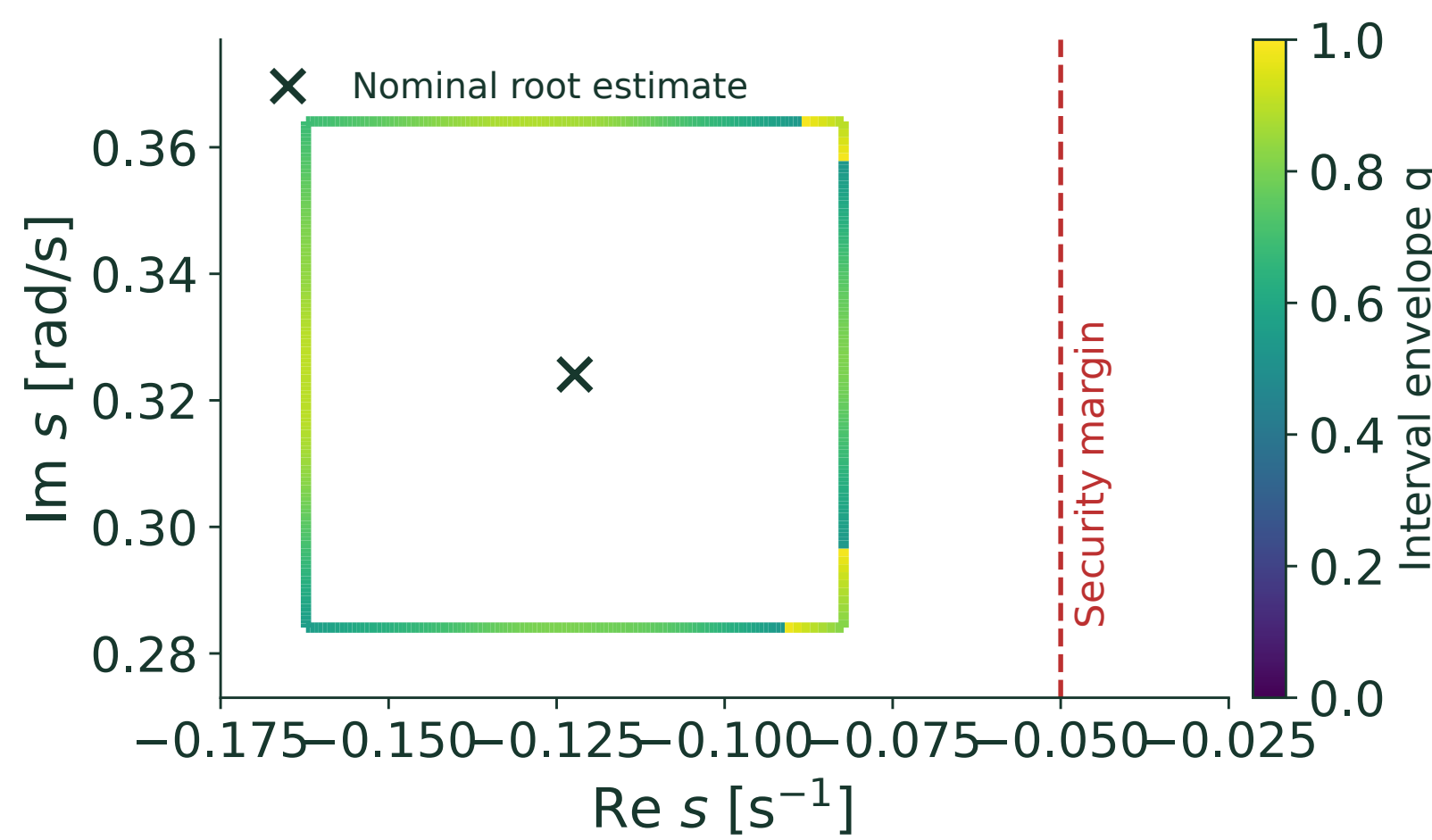
Full retained-device DAE reduction; true delay by method of steps, Rodas5P; 60 s, tolerance 10^{-9} . All 39 bus phase frequencies.



$$|\Delta f|_{\max} = 0.4746 \text{ Hz} \quad |\text{RoCoF}|_{\max} = 0.2008 \text{ Hz/s}$$

Dashed limits: 0.5 Hz and 0.5 Hz/s. Both metrics use a 0.5 s phase window. Frozen impedance-load events at the indicated nominal MW.

8 A CONTINUOUS REGIONAL CERTIFICATE



Every point in the box

$$\begin{aligned} |\delta \rho_i| &\leq 0.001 \\ |\delta K_i|/K_i^0 &\leq 1\% \end{aligned}$$

for both K_p, K_i ; center $\rho_i^0 = 0.875$.

Exactly one root

$\Re \lambda < -0.08229$
inside the contour for every parameter vector in the box.

485 interval panels; 128-bit complex ball arithmetic; envelope $q < 0.98794235 < 1$.

One exported-model root cluster, not the complete spectrum. The certificate encloses the whole contour and parameter box at exact $\tau = 1/25$ s; exported binary64 coefficients are treated as exact data.

9 WHAT IS ESTABLISHED – AND WHAT REMAINS OPEN

0.972–1.079 pu

voltage range; limits [0.9, 1.1]

0.00634

minimum SG slack; limit 0.002

Feasibility shown

all-root margin checked numerically

Maximum still open

no certified replacement ceiling

Same- ρ comparison: fixed gains also pass all five events. Retuning slightly lowers RoCoF but increases frequency excursion; superiority is not established. No converter current-limit claim.

TAKEAWAY

LOCAL DELAY. TWO OPPOSITE DAMPING EFFECTS.

Network paths determine the sign; collective bounds quantify the interaction.

REFERENCES

- [1] Marković et al., IEEE TPWRS, 2021.
[2] Bindel & Hood, SIAM J. Matrix Anal., 2013.
[3] Beyn, Linear Algebra Appl., 2012.

- [4] Johansson, IEEE Trans. Comput., 2017.
Theory, proofs, full gain vectors and data: QR.

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